

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

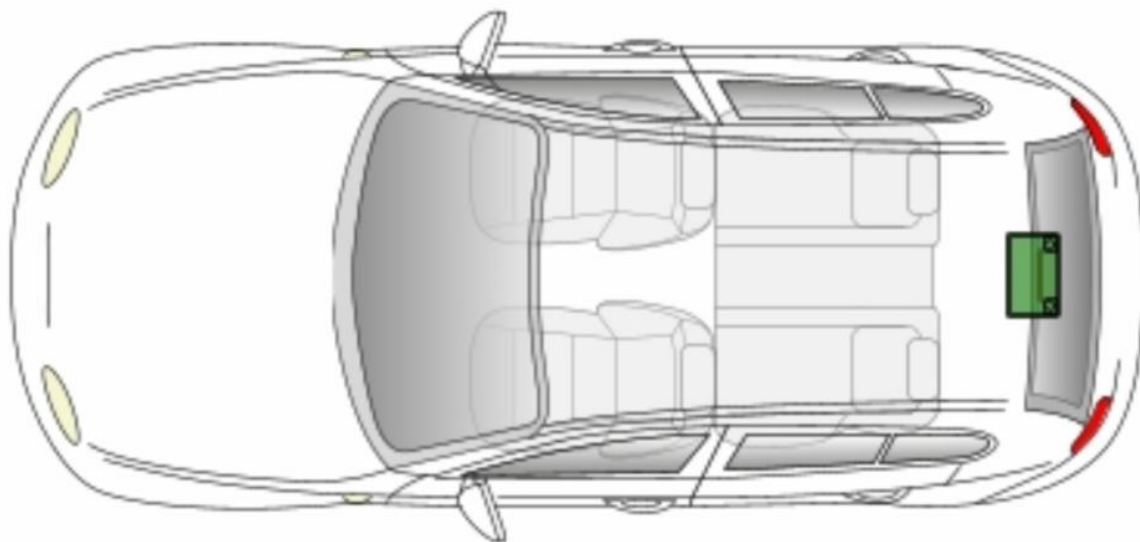
Note: Avoid bending, torsion or traction

Always use new gaskets

General

Before disconnecting the battery cable, check the audio system security code

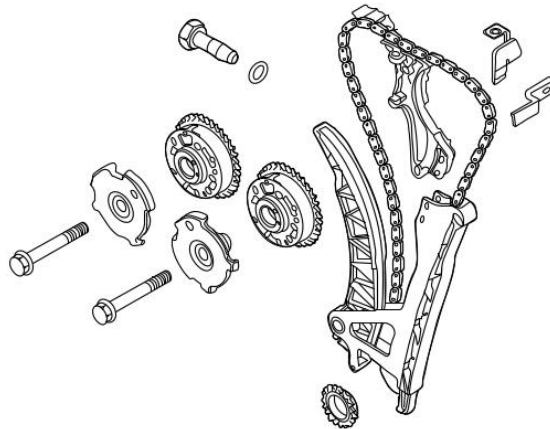
Battery location



Luggage compartment

Timing chain layout

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Removal

Disconnect the battery earth cable

Remove the mass airflow meter

Remove the air filter housing

Remove the ignition coil cover

Remove the ignition coils

Disconnect the wiring loom from the valve cover

Caution! The fuel system is most likely pressurised

Wear hand and eye protection when working on the fuel system

Release the fuel pressure

Remove the high-pressure fuel lines between the rail and the injectors

Open fuel connections must be sealed off with appropriate covers or plugs (32 1 270)

Remove the fuel line between the high-pressure pump and the fuel rail

Remove the high-pressure pump

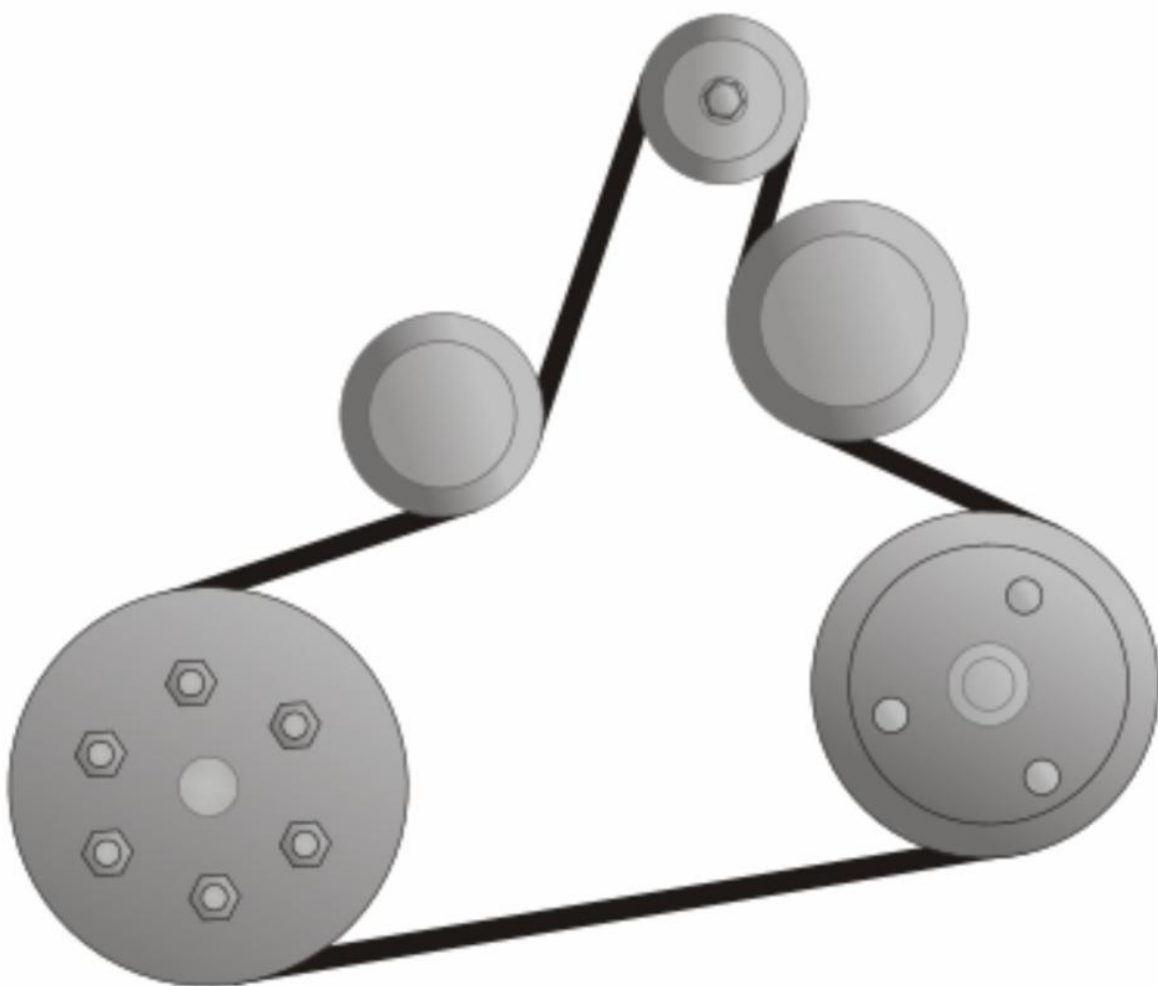
Disconnect the crankcase breather hose from the valve cover

Remove the valve cover

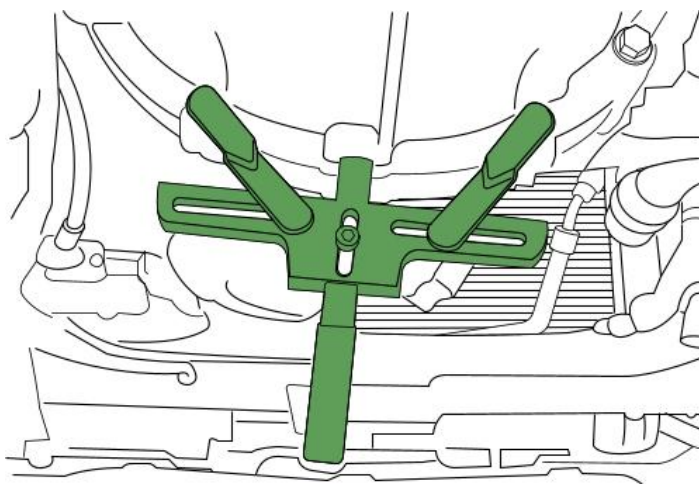
Remove the ancillary drive belt

Drive belt layout

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Fit the crankshaft locking tool (11 8 180)



Remove the crankshaft pulley

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Remove the crankshaft locking tool (11 8 180)

Place the bonnet in the service position

Attach the engine to the engine support bar (00 6 001, 00 6 031, 00 6 051, 00 6 052, 00 6 053)

Remove the engine supports nuts

Lift the engine slightly ((10 mm))

Remove the reinforcement plate

Remove the underbody covers

If applicable: Detach the brackets from the left and right control arms of the engine subframe.

Lower the subframe

If required: Remove the steering box cover.

Detach the steering column from the steering rack

Place the jack under the front axle subframe

Fit a special tool to support the subframe (31 5 250, 00 2 040)

Remove the subframe bolts

Lower the subframe ((< 10 cm))

Remove the front splash guard

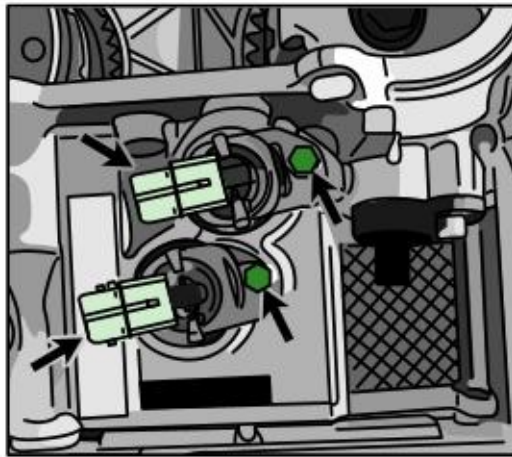
Drain the engine oil

Disconnect the oil level sensor

Remove the oil sump

Disconnect the electrical connectors

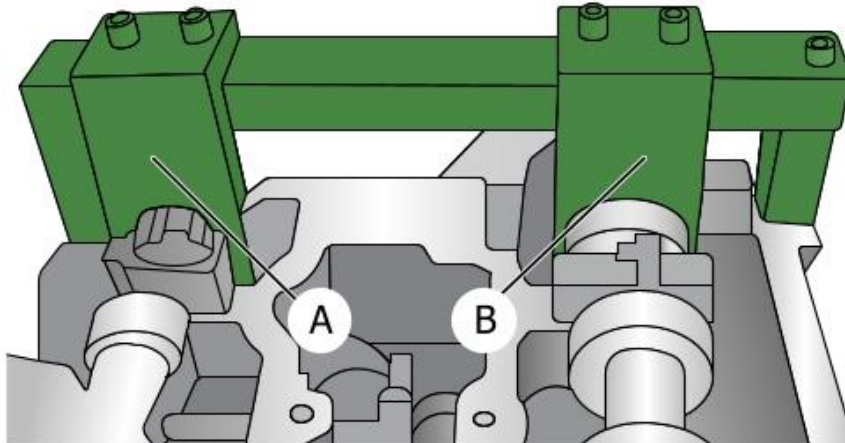
Remove the valve timing control solenoid valves



Turn the crankshaft to TDC for cylinder 1

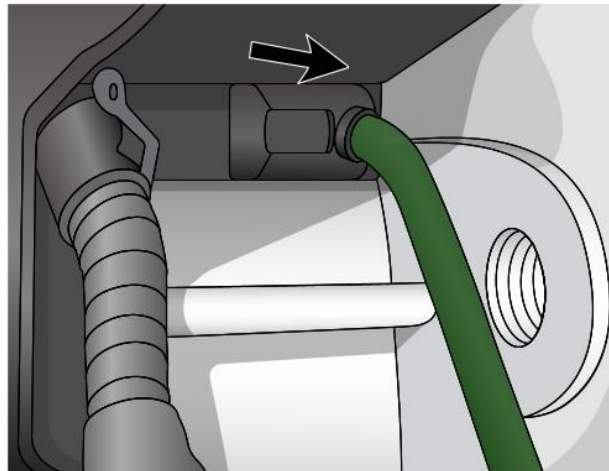
Fit the camshaft locking tools ((A-B)) (11 8 691, 11 8 692, 11 8 693)

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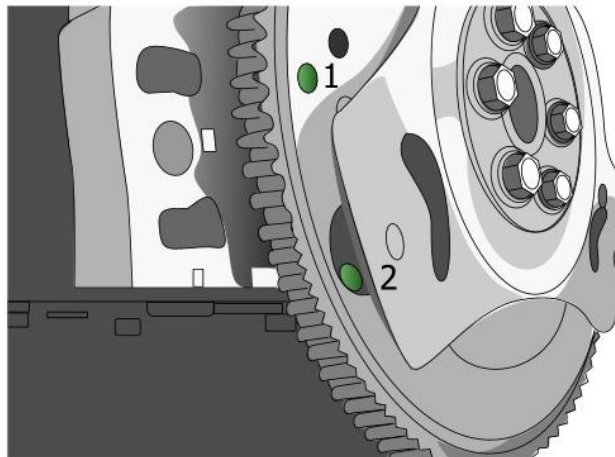


Check that cylinder 1 is still at TDC

Fit the flywheel locking tool (11 5 120)



Vehicles with automatic transmission: The special tool must be secured in bore No. 1.



Remove the spark plugs

Loosen the camshaft gearwheel (inlet)

Loosen the camshaft gearwheel (exhaust)

Remove the chain tensioner piston

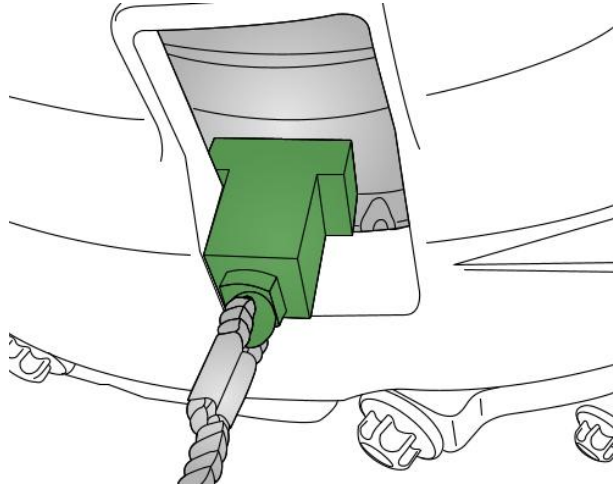
Remove the inlet camshaft timing adjuster

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Remove the exhaust camshaft timing adjuster

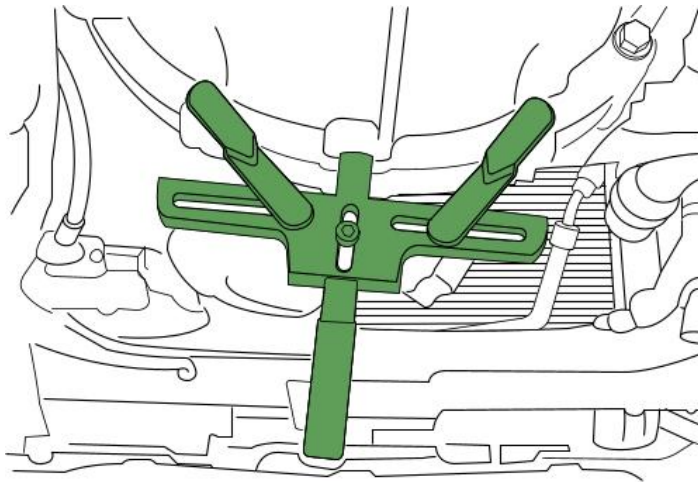
Vehicles with automatic transmission:

Fit the crankshaft locking tool (11 8 660)

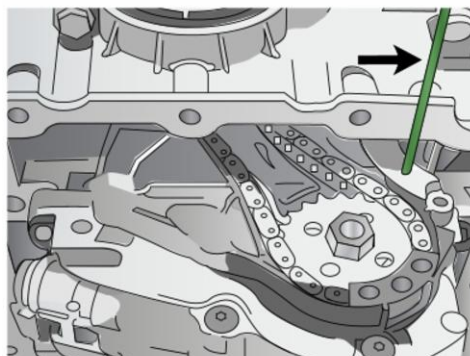


Vehicles with manual transmission:

Fit the crankshaft locking tool (11 8 180)



Lock the chain tensioner with a locking pin (11 3 340)



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All vehicles:

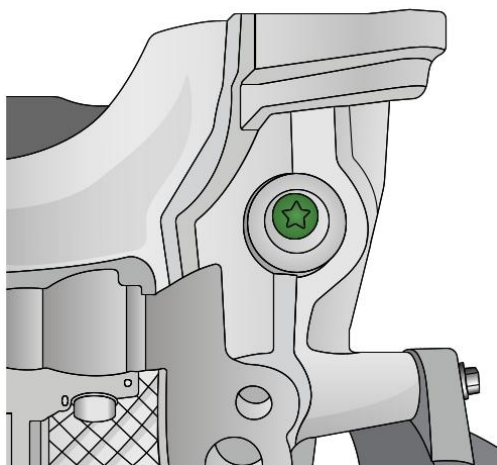
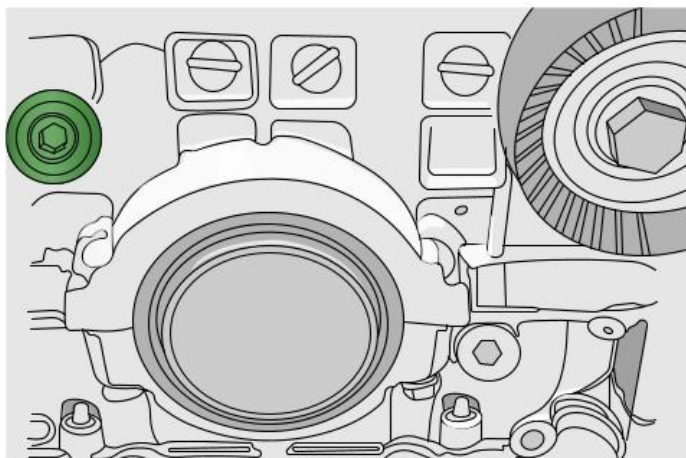
Check that the special tool is correctly fitted (11 5 120)

Check that the camshaft locking tool is in position (11 8 691, 11 8 692, 11 8 693)

Loosen the crankshaft gearwheel bolt

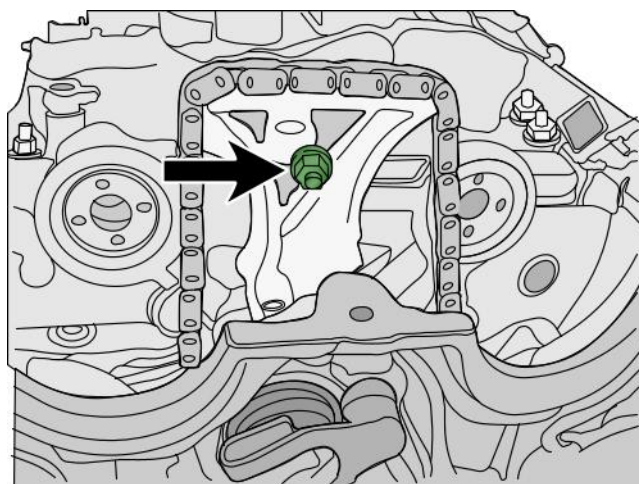
Remove the crankshaft hub

Remove the access plugs



Remove the bearing pins for the chain guide rail and the tensioner rail

Remove the bearing pin from the cylinder head



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Remove the chain guide with the timing chain, guide rail and tensioner rail

Remove the oil pump tube

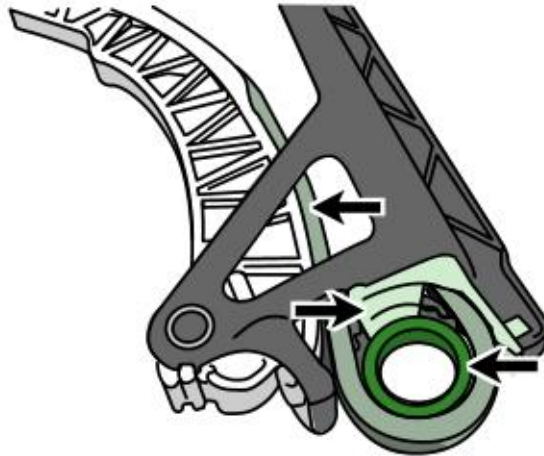
Installation

Refit the crankshaft gearwheel

Refit the chain guide with the timing chain, guide rail and tensioner rail

Tension the chain

Note: Ensure correct fitting



Refit the timing chain guide bolts to the cylinder head

Refit the bearing pins for the chain guide rail and the tensioner rail

Refit the access plugs

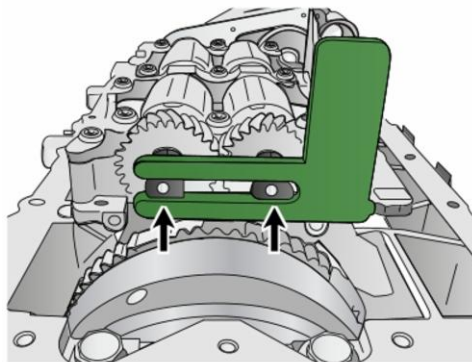
Refit the crankshaft hub

Fit the centre bolt of the crankshaft gearwheel/vibration damper

Note: Finger-tight only

Align the balance shafts

Use the special tool (11 8 700)



Tighten the crankshaft gearwheel bolt

Remove the special tool (11 8 700)

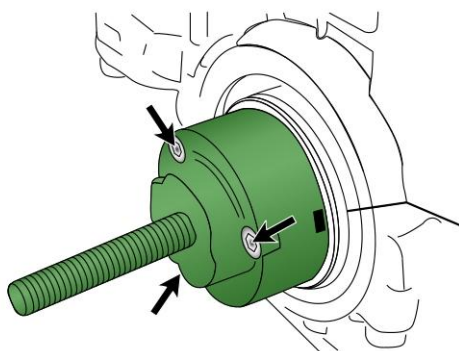
Remove the crankshaft locking tool (11 8 180, 11 8 660)

Refit the oil pump tube

Renew the crankshaft front radial seal

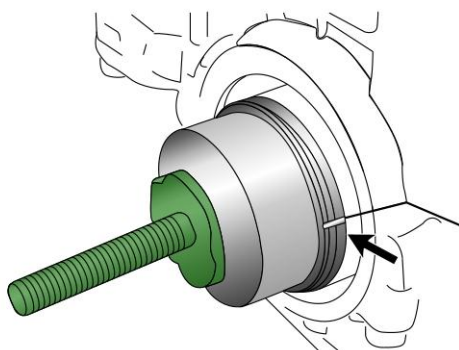
Note: Clean off the sealing surfaces

Fit the special tool (11 9 232)

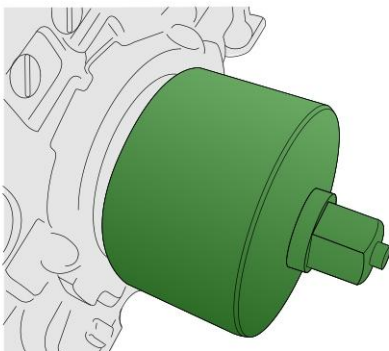
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Fit the crankshaft seal over the special tool

Align the marks



Assemble the special tool as shown (11 9 231, 11 9 232, 11 9 233, 11 9 234)

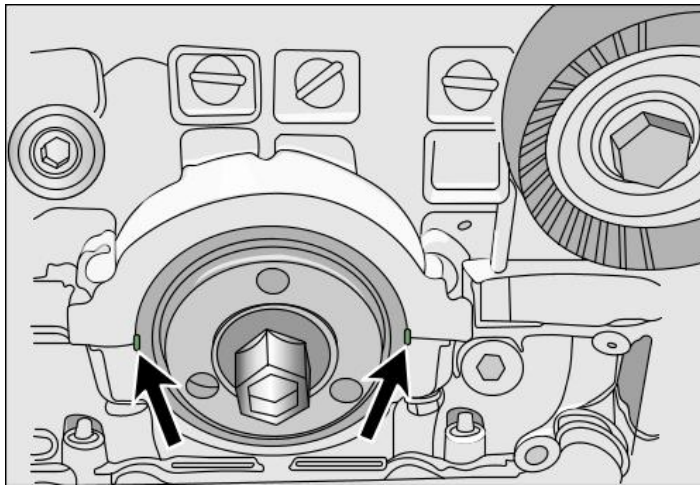


Tighten the special tool bolt until full surface contact is made

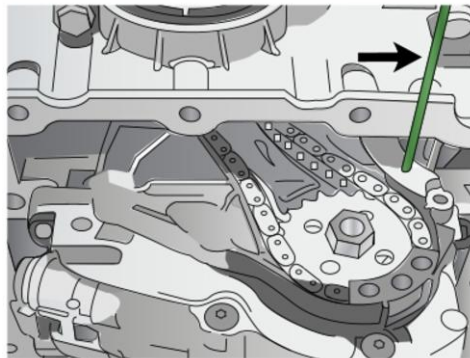
Remove the special tools (11 9 231, 11 9 232, 11 9 233, 11 9 234, 11 9 232)

Apply sealant as indicated. Loctite: ((171000, 193140)).

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Remove the tensioner locking pin (11 3 340)



Check the camshaft gearwheel markings

Exhaust camshaft gearwheel: ((AUS/EX))

Inlet camshaft gearwheel: ((EIN/IN))

Renew the camshaft gearwheel bolts

Pull the timing chain upwards

Tension the chain

Fit the timing chain onto the inlet camshaft gearwheel

Refit the camshaft sensor gear

Tighten the camshaft gearwheel bolt by hand

Pull the timing chain upwards

Tension the chain

Fit the timing chain onto the exhaust camshaft gearwheel

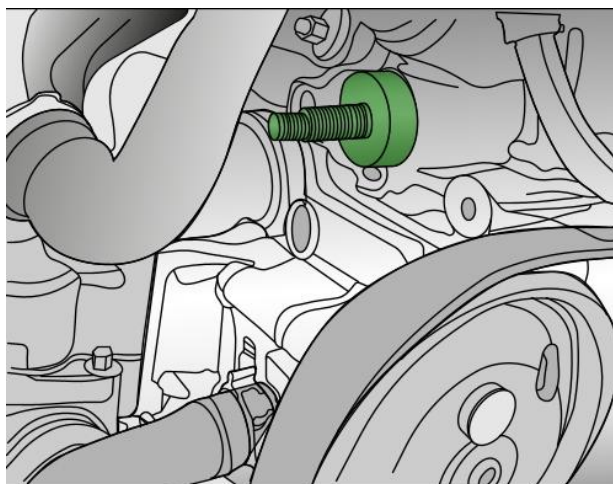
Refit the camshaft sensor gear

Tighten the camshaft gearwheel bolt by hand

Press down on the tensioner rail

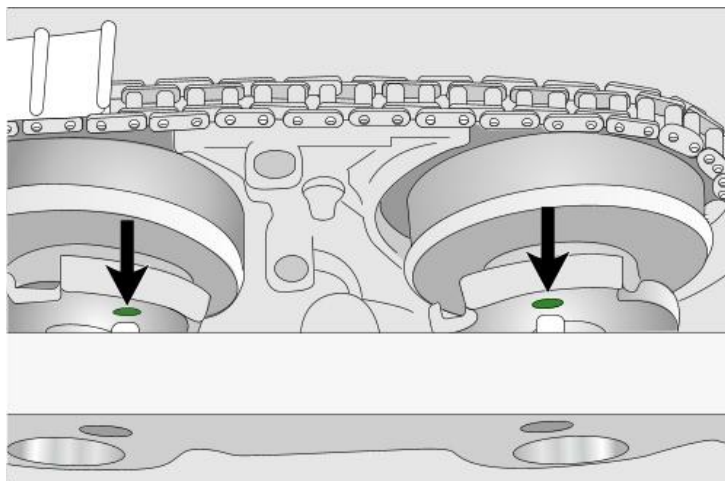
Fit the dummy chain tensioner (11 9 340)

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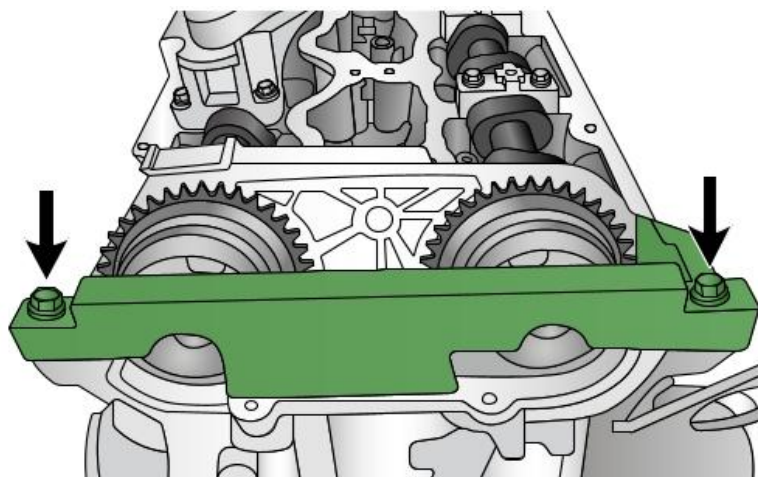


Fit the alignment tool (11 8 711, 11 8 712)

Align the camshaft gearwheel marks as shown



Secure the special tool with two bolts

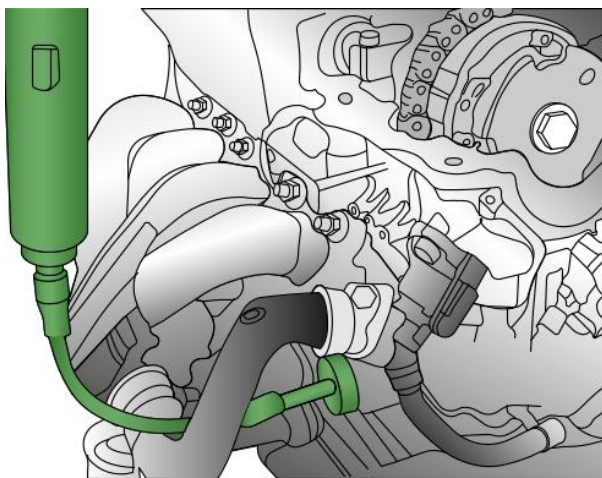


Loosen the camshaft gearwheel bolts ((180°))

Tighten the camshaft gearwheel bolts lightly until there is no play

Pre-tension the timing chain

Use the special tool ((0.6 Nm)) (00 9 250)

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Tighten the camshaft gearwheel bolts (00 9 120)

Remove the special tools (11 8 711, 11 8 712, 11 9 340)

Compress the chain tensioner piston twice to drain the oil

Refit the chain tensioner piston

Remove the flywheel locking tool (11 9 190)

Turn the crankshaft clockwise 2 rotations

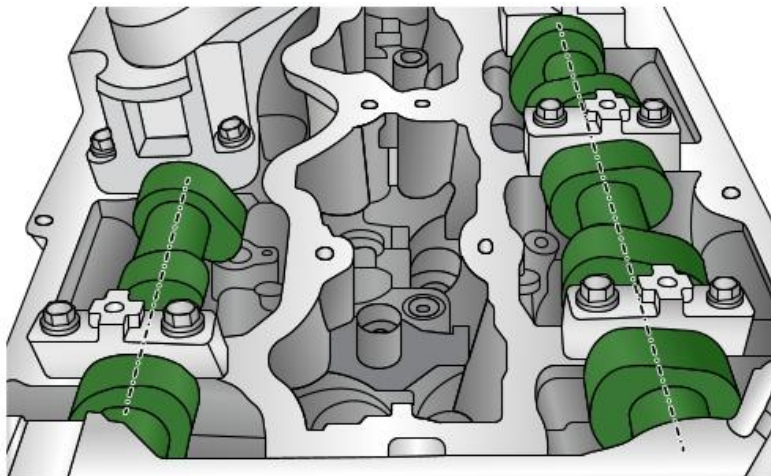
Check the valve timing

Proceed as follows:

Turn the crankshaft clockwise until cylinder 1 is at TDC

Note: Cylinder 1 cams at TDC firing position

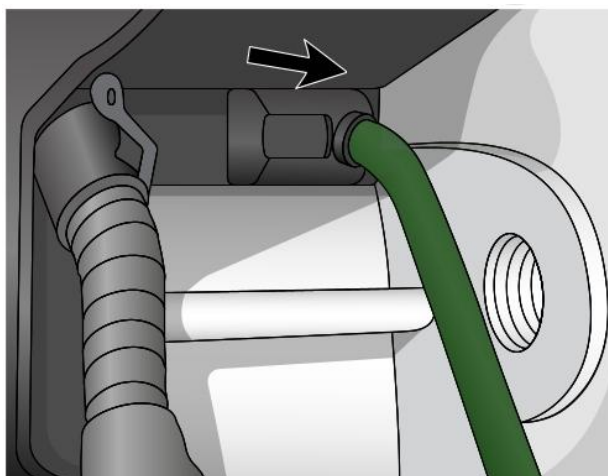
Note: The cams for cylinder 1 must point upwards at an angle, facing each other



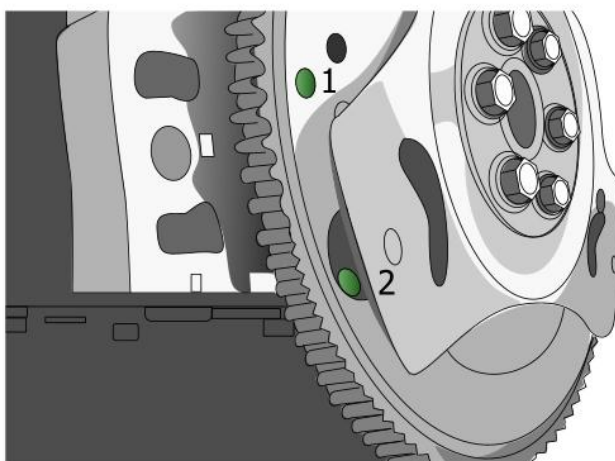
Fit the flywheel locking tool. Through the access hole. Near the starter motor.

Note: Ensure correct fitting

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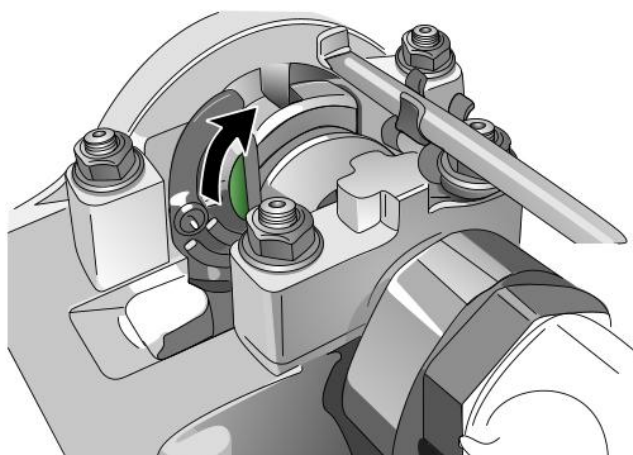


Vehicles with automatic transmission: The special tool must be secured in bore No. 1.



Check the camshaft adjusters

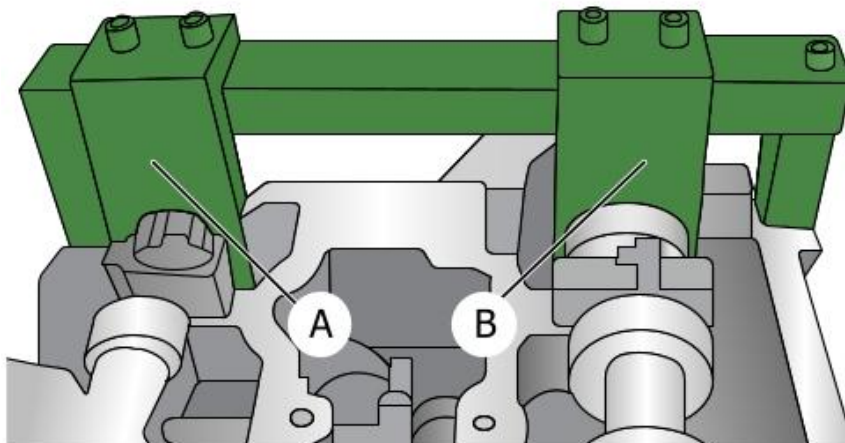
Turn the camshafts to check that the camshaft adjustment units can be locked



Action required if a fault is found: Renew the camshaft adjuster.

Assemble the special tool as shown (11 8 691, 11 8 692, 11 8 693)

If the camshaft timing is correct, the special tool will touch the cylinder head on both sides ((A-B))

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Inlet side: Maximum protrusion: ((0.5 - 1.0 mm)).

If the camshaft locking tool does not fit, adjust the timing

Proceed as follows:

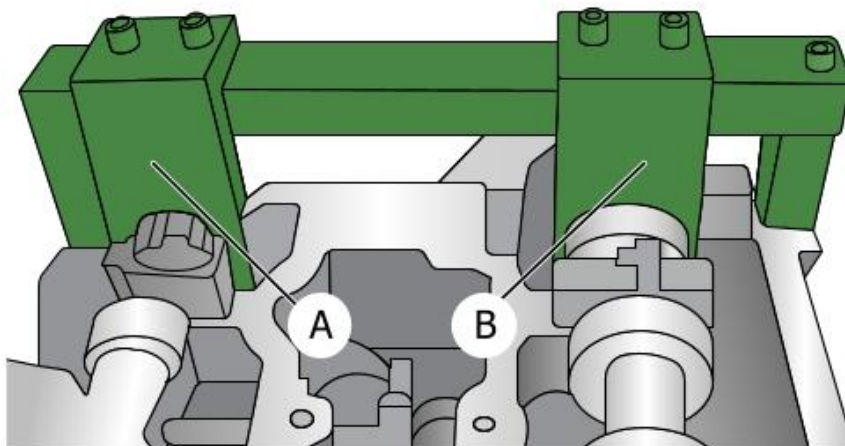
Check that the flywheel locking tool is correctly fitted (11 9 190)

Loosen the camshaft gearwheel bolts

Tighten the camshaft gearwheel bolts lightly until there is no play

Fit the camshaft locking tools (11 8 691, 11 8 692, 11 8 693)

If required: Turn the camshafts until the locking tools fit ((A-B)).



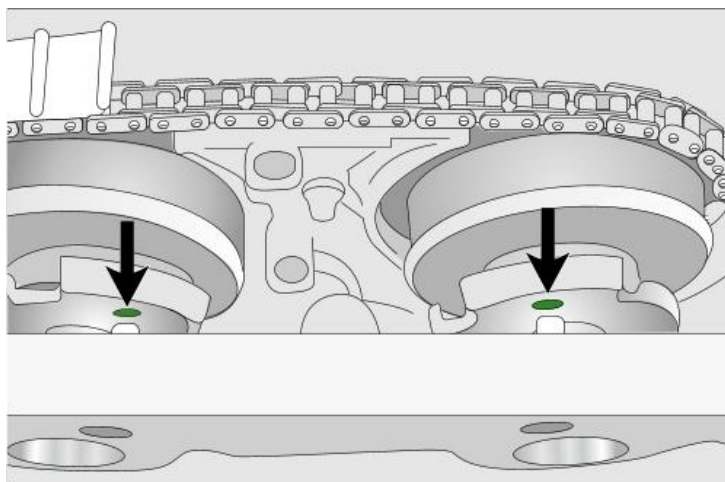
Remove the chain tensioner

Fit the dummy chain tensioner (11 9 340)

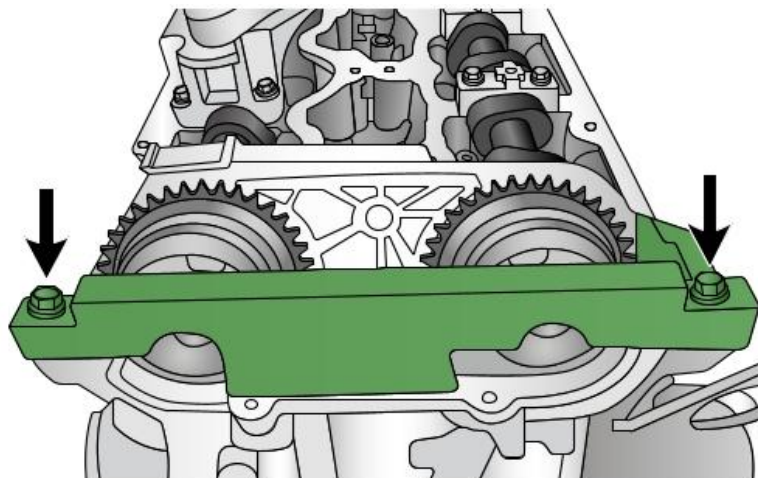
Fit the alignment tool (11 8 711, 11 8 712)

Align the camshaft gearwheel marks as shown

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Secure the special tool with two bolts



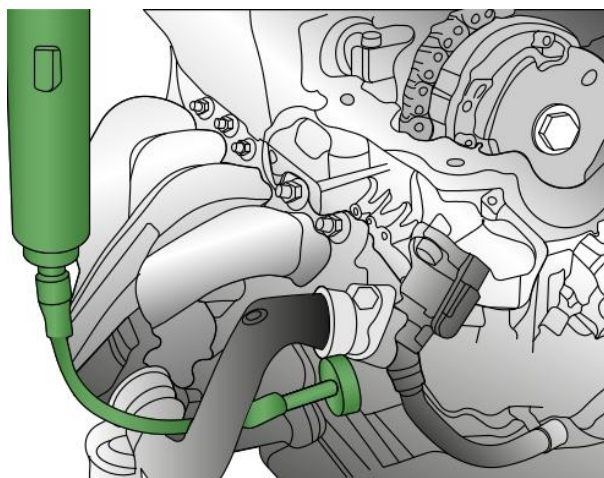
Renew the camshaft gearwheel bolts

Tighten the camshaft gearwheel bolts lightly until there is no play

Loosen the camshaft gearwheel bolts ((180°))

Pre-tension the timing chain

Use the special tool ((0.6 Nm)) (00 9 250)



Tighten the camshaft gearwheel bolts

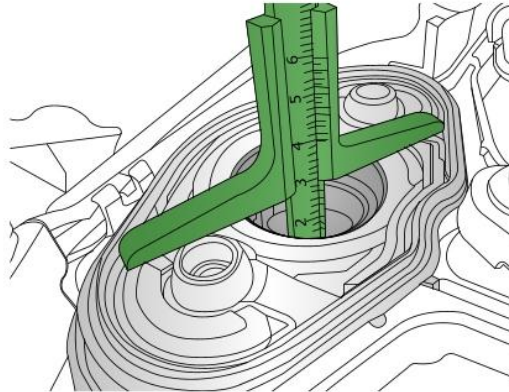
Remove all the locking tools (11 5 120, 11 8 691, 11 8 692, 11 8 693)

Turn the crankshaft clockwise 2 rotations

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Turn the crankshaft to TDC for cylinder 1
 Check the valve timing again
 Further refitting is in the reverse order of removal

High-pressure fuel pump:
 Renew the seal
 Fit the gauge



Slowly turn the crankshaft clockwise
 Check the gauge reading
 When the maximum value is reached: Refit the high-pressure pump.

Tighten the bolts in 1/4 turns

Refill the engine oil
 Engine sump, including filter 4.25 (l)
 Reconnect the battery earth cable
 Connect the diagnostic tool
 Check for any fault codes

Vehicles with active steering:
 Calibrate the steering angle sensor

Torque settings

Timing chain tensioner

Timing chain tensioner 65 (Nm)

Renew the seal

Timing chain guide

Timing chain guide (M7) 14 (Nm)

Timing chain guide (M8) 20 (Nm)

Camshaft gearwheel

Camshaft gearwheel

Renew the bolts

Stage 1 20 (Nm)

Stage 2 180 (°)

Crankshaft gearwheel

Crankshaft gearwheel (Centre bolt)

Renew the bolt (Threads lightly oiled)

Stage 1 50 (Nm)

Stage 2 150 (°)

Oil sump

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Sump 30 (Nm)

Engine oil drain plug

Engine oil drain plug 25 (Nm)

Crankshaft pulley

Crankshaft pulley (Outer bolts) 35 (Nm)

Spark plugs

Spark plugs (Caution: only use a flexible bi-hex socket (M12); damage may occur if standard tools are used) 20 - 26 (Nm)

Valve cover

Valve cover 10 (Nm)

Suspension: Refer to 'Technical drawings'

Steering: Refer to 'Technical drawings'

Access plug: (M16 x 1.5: 32 Nm)

Access plug: (M18 x 1.5: 25 Nm)

Camshaft timing solenoid valve: (10 Nm)

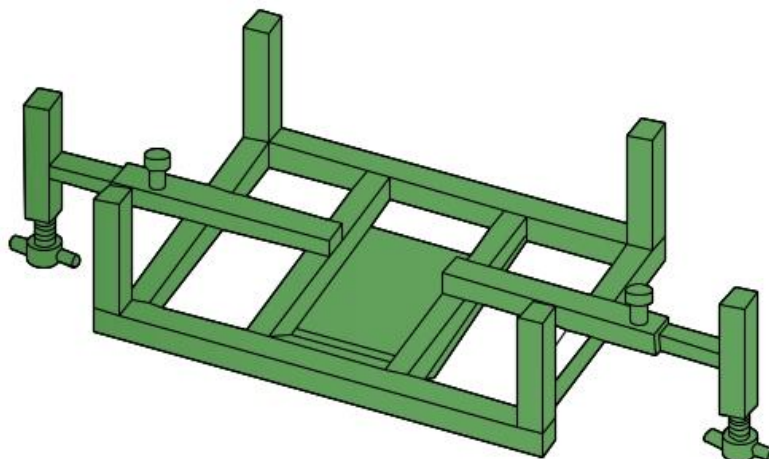
Reinforcement brace:

Stage 1: (56 Nm)

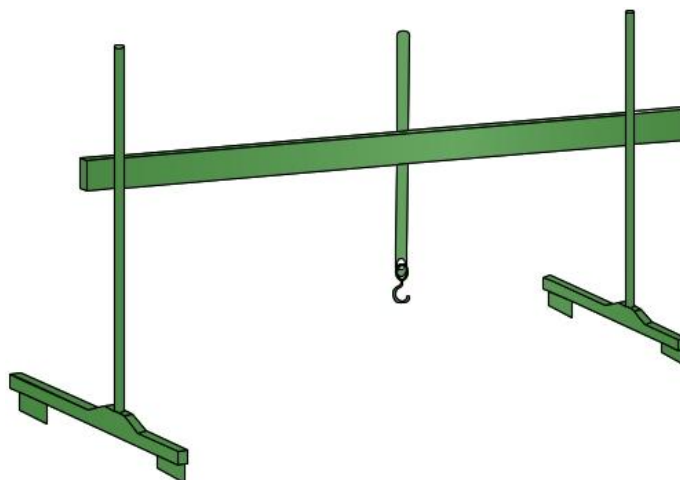
Stage 2: (90°)

Special tools

Support bracket: (00 2 040)

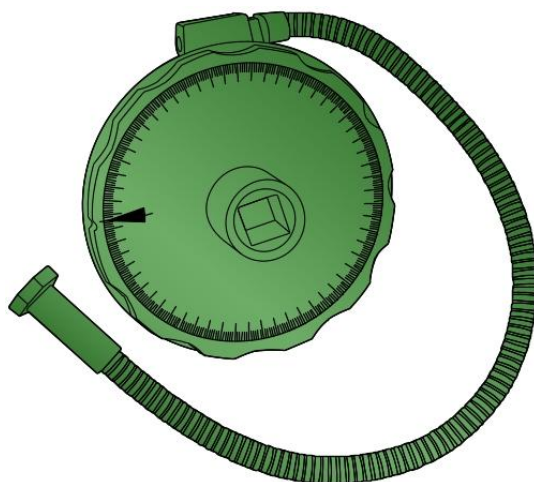


Engine support tool: (00 6 001, 00 6 031, 00 6 051, 00 6 052, 00 6 053)

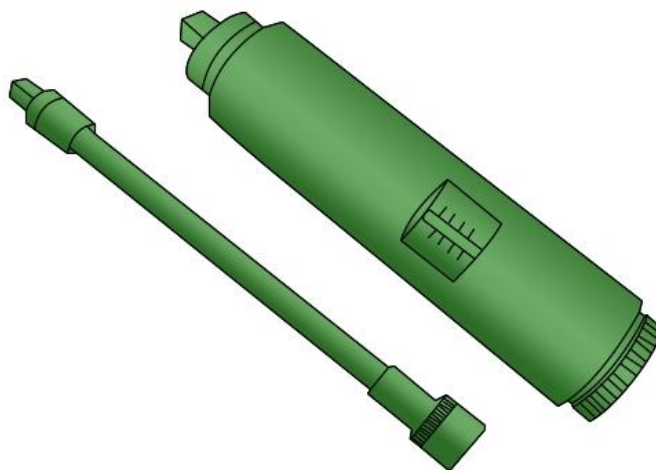


Torque wrench: (00 9 120)

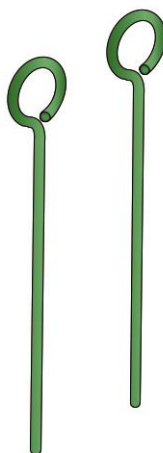
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Adjustment tool: (00 9 250)

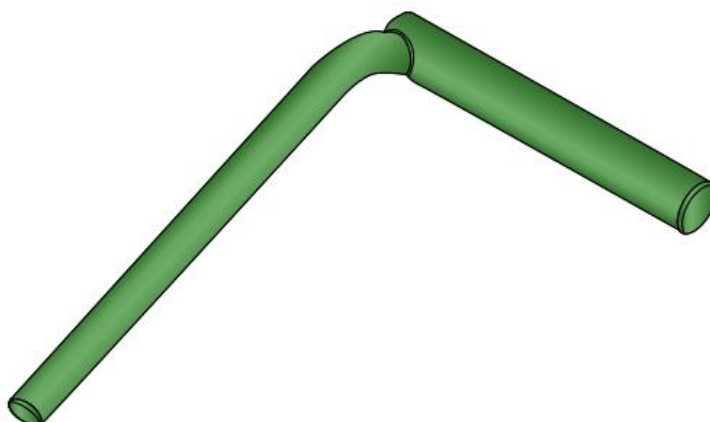


Chain tensioner locking pin: (11 3 340)

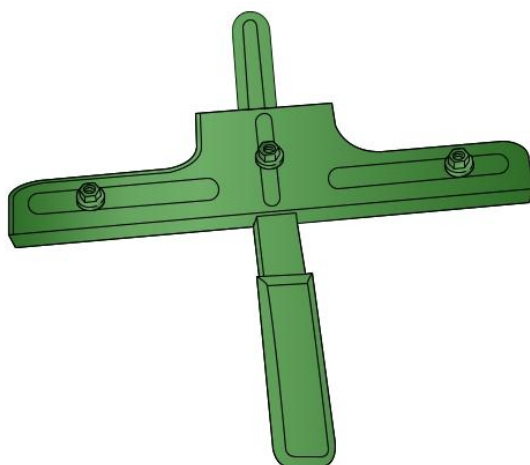


Flywheel locking tool: (11 5 120)

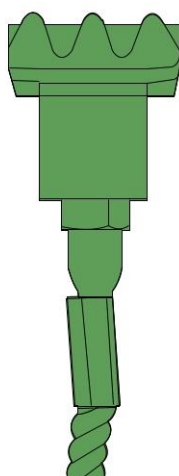
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Crankshaft locking tool: (11 8 180)

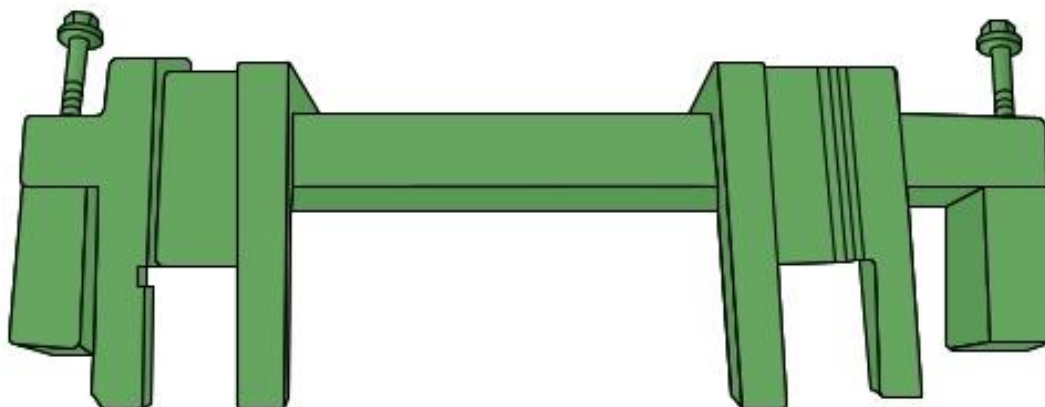


Flywheel locking tool: (11 8 660)

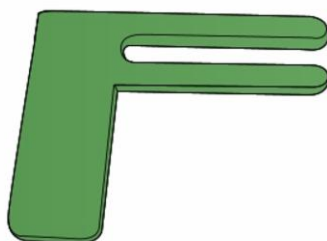


Camshaft timing tool: (11 8 691, 11 8 692, 11 8 693)

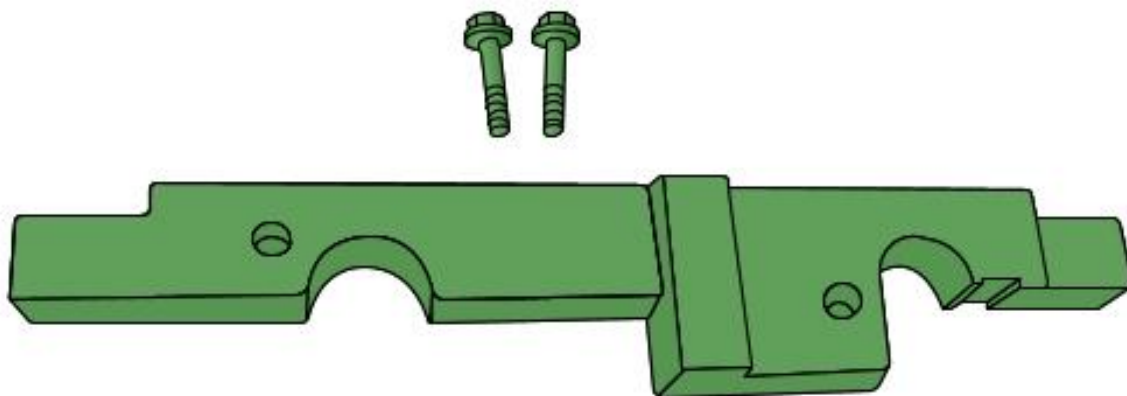
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Alignment tool: (11 8 700)

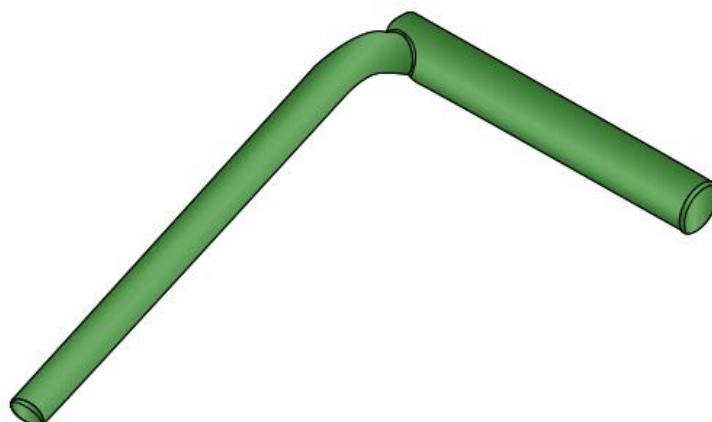


Camshaft alignment tool: (11 8 711, 11 8 712)

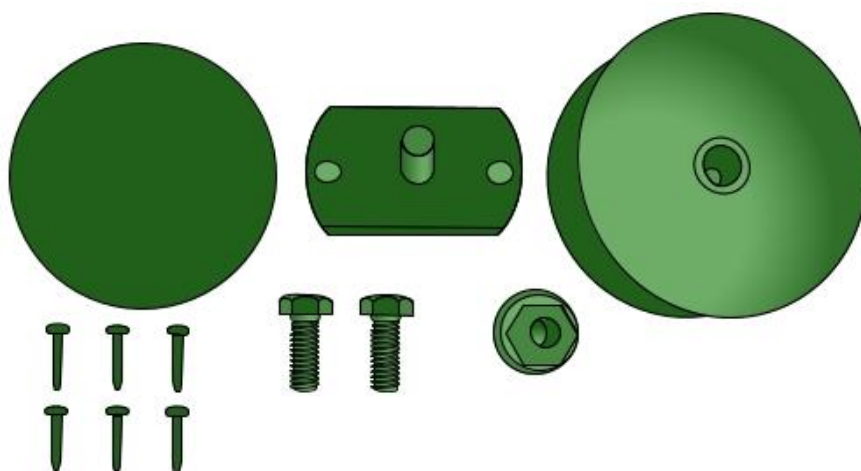


Crankshaft locking tool: (11 9 190)

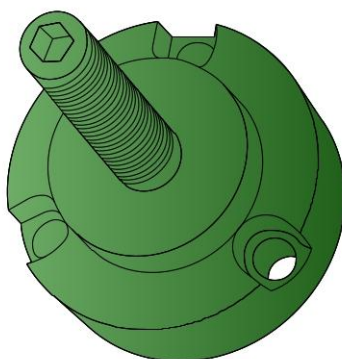
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Crankshaft front radial seal fitting tool: (11 9 231, 11 9 232, 11 9 233, 11 9 234)

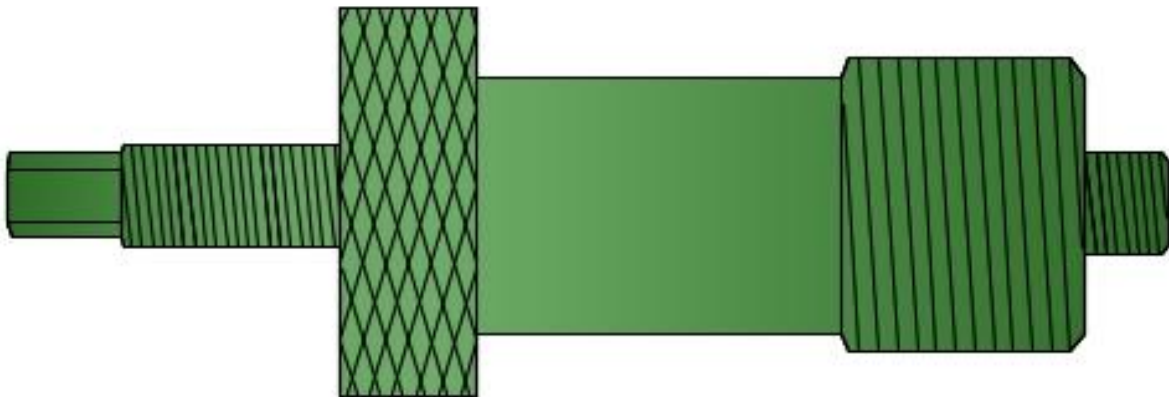


Oil seal fitting tool: (11 9 232)

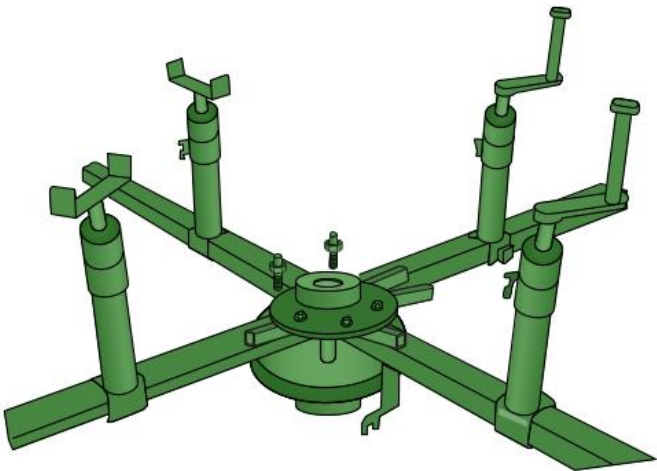


Adaptor: (11 9 340)

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Subframe support: (31 5 250)



Plugs kit: (32 1 270)

